PROJECT PROPOSAL

ONLINE LEARNING PLATFORM

Database Design and System Overview

Project Information	Details
Project Type	Web-based Learning Management System
Database Name	online_learning
Database Server	MariaDB / MySQL
Development Technology	PHP, MySQL, HTML, CSS and JavaScript
Reference Application	SkillSet – Online Learning Platform

Prepared for Academic Project Presentation

SUBMITTED BY	STUDENT DETAILS
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1. Introduction

The Online Learning Platform is a web-based application designed to make online education easier and more organized. Students can explore courses, enroll in courses, watch lessons, attend live classes, take quizzes and receive certificates. Instructors can publish courses and manage learning materials, while administrators can manage users and platform activities.

This project is planned with reference to the general features of the SkillSet Online Learning Platform. The database structure documented here is based on the SQL file supplied for this project.

2. Project Objectives

- Provide a single platform for online learning.
- Allow students to register, log in and manage their learning activities.
- Allow instructors to create courses and add sections and lessons.
- Support course enrollment and payment records.
- Provide quizzes, questions, answer options and quiz results.
- Support live class schedules, course reviews, wishlists and certificates.
- Organize information through a relational MySQL database.

3. Main User Roles

User Role	Main Responsibilities
Administrator	Manages users, courses, categories and overall platform information.
Instructor	Creates courses, adds course content, schedules live classes and supports learners.
Student	Browses courses, enrolls, makes payments, studies lessons, takes quizzes and reviews courses.

4. Main Features of the System

Feature	Description
User Management	Registration, login, user roles and account status.
Course Management	Course title, description, price, thumbnail, category and publication status.
Course Content	Sections and lessons containing video links and learning content.
Enrollment	Records which student has enrolled in which course.
Payment Management	Stores payment amount, method, transaction ID and payment status.
Quiz System	Stores quizzes, questions, answer options and correct-answer indicators.
Quiz Results	Stores student quiz attempts and obtained marks.
Live Classes	Stores class title, meeting link and scheduled start time.
Certificates	Stores certificate numbers and issue dates for students.
Reviews and Wishlist	Allows students to review courses and save courses for later.

5. Database Overview

The supplied SQL database is named online_learning. It contains 15 tables. The tables are connected through primary keys and foreign keys so that related information can be stored consistently.

No.	Table Name	Purpose
1	users	Stores administrator, instructor and student accounts.
2	categories	Stores course categories.
3	courses	Stores course details and links each course to an instructor and category.
4	course_sections	Stores the sections or chapters of each course.
5	lessons	Stores lesson titles, video URLs and lesson content.
6	enrollments	Stores student course enrollment records.
7	payments	Stores student payments for courses.
8	quizzes	Stores quizzes associated with courses.
9	questions	Stores questions belonging to quizzes.
10	quiz_options	Stores answer options and indicates the correct option.
11	quiz_attempts	Stores student quiz attempts and obtained marks.
12	certificates	Stores certificate details for students and courses.
13	live_classes	Stores live class details and meeting links.
14	course_reviews	Stores student ratings and written reviews for courses.
15	wishlists	Stores courses saved by students.

6. Important Database Tables Explained

Table	Important Fields	Explanation
users	id (PK), name, email, password, phone, role, status, created_at	Stores all platform accounts. The role field identifies admin, instructor or student.
categories	id (PK), category_name, description, status	Groups courses into categories such as Programming, Design or Business.
courses	id (PK), instructor_id (FK), category_id (FK), title, description, price, thumbnail, status	Stores course information and identifies its instructor and category.
course_sections	id (PK), course_id (FK), section_title	Divides a course into chapters or sections.
lessons	id (PK), section_id (FK), lesson_title, video_url, content	Stores learning materials under a course section.
enrollments	id (PK), student_id (FK), course_id (FK), enrollment_date, status	Connects students with the courses they join.
payments	id (PK), student_id (FK), course_id (FK), amount, payment_method, transaction_id, status	Keeps a record of course payment information.
quizzes	id (PK), course_id (FK), title, total_marks, pass_marks	Stores assessments for a course.
questions	id (PK), quiz_id (FK), question, marks	Stores individual questions in a quiz.
quiz_options	id (PK), question_id (FK), option_text, is_correct	Stores possible answers for each question.
quiz_attempts	id (PK), student_id (FK), quiz_id (FK), obtained_marks, attempt_date	Records a student's quiz attempt and score.
certificates	id (PK), student_id (FK), course_id (FK), certificate_number, issue_date	Stores certificate information issued to students.
live_classes	id (PK), course_id (FK), instructor_id (FK), class_title, meeting_url, start_time	Stores scheduled online classes.

course_reviews	id (PK), student_id (FK), course_id (FK), rating, review	Stores student feedback about courses.
wishlists	id (PK), student_id (FK), course_id (FK)	Stores courses that students want to save.

7. Primary Key and Foreign Key

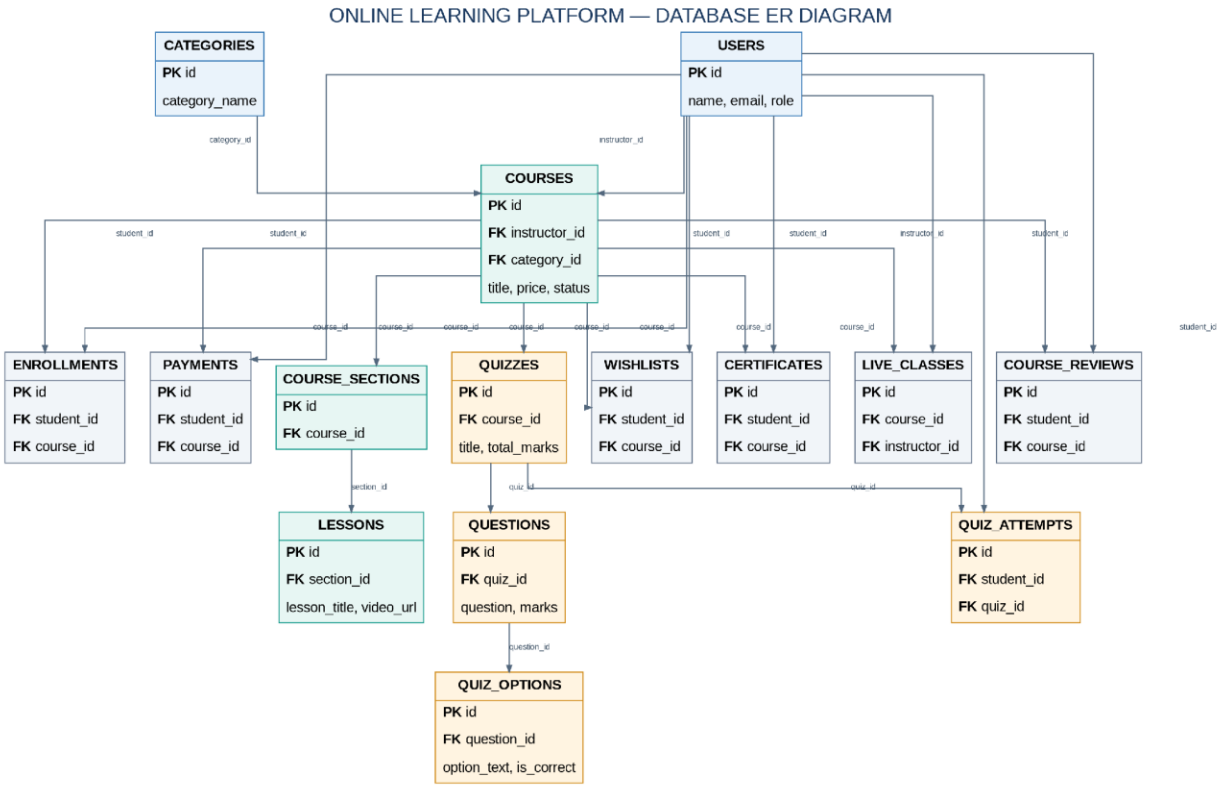
A Primary Key (PK) uniquely identifies each record in a table. A Foreign Key (FK) links a record to a related record in another table.

Key Type	Example from this database	Purpose
Primary Key (PK)	users.id	Uniquely identifies a user.
Foreign Key (FK)	courses.instructor_id	Links a course to its instructor in users.id.
Foreign Key (FK)	courses.category_id	Links a course to categories.id.
Foreign Key (FK)	enrollments.student_id	Links an enrollment to users.id.
Foreign Key (FK)	enrollments.course_id	Links an enrollment to courses.id.

8. Database Relationships

Parent Table(s)	Related Table	Relationship Explanation
users	courses	One instructor can create many courses.
categories	courses	One category can contain many courses.
courses	course_sections	One course can contain many sections.
course_sections	lessons	One section can contain many lessons.
users	enrollments	One student can have many enrollment records.
courses	enrollments	One course can have many enrolled students.
users and courses	payments	Each payment record is linked to a student and a course.
courses	quizzes	One course can contain many quizzes.
quizzes	questions	One quiz can contain many questions.
questions	quiz_options	One question can have multiple answer options.
users and quizzes	quiz_attempts	A student can make quiz attempts.
users and courses	certificates	A certificate is linked to a student and a course.
courses and users	live_classes	A live class is linked to a course and an instructor.
users and courses	course_reviews	A student can review a course.
users and courses	wishlists	A student can save courses to a wishlist.

9. Simplified ER Diagram



PK = Primary Key | FK = Foreign Key

10. System Workflow

1. A user registers and logs in to the platform.
2. The instructor creates a course and selects a category.
3. The instructor adds course sections and lessons.
4. The student browses available courses and selects a course.
5. The student enrolls and the payment information is recorded.
6. The student studies lessons and can join scheduled live classes.
7. The student takes quizzes; questions, options and marks are stored.
8. The platform can store course reviews and certificate details.

11. Technology Used

Technology	Use in the Project
HTML	Creates the structure of web pages.
CSS	Controls the design and layout.
JavaScript	Adds client-side interaction.
PHP	Handles server-side logic and database operations.
MySQL / MariaDB	Stores and manages application data.
Laragon / XAMPP	Provides a local development environment.
phpMyAdmin	Helps create and manage the database.

12. Expected Benefits

- Students can access course materials from one platform.
- Instructors can organize and publish learning content.
- Course enrollment and payment records are stored systematically.
- Quizzes and results help track student learning.
- Live class links, reviews, wishlists and certificates improve the learning experience.
- A relational database helps maintain organized and connected information.

13. Conclusion

The Online Learning Platform is designed to bring students, instructors and administrators together in one system. Its 15-table database supports user accounts, courses, lessons, enrollment, payments, quizzes, live classes, certificates, reviews and wishlists. This database provides a foundation for developing the application using PHP and MySQL.

14. Future Improvements

- Add lesson progress tracking and course completion rules.
- Add more detailed payment and refund management.
- Add notifications and messaging between students and instructors.
- Add downloadable learning resources and attendance tracking.
- Improve reporting and analytics for administrators.

15. Short Presentation Script

Good morning, respected sir. My name is Farjana Akter. My student ID is 1297023 and my batch is PWAD 73. Today I am going to present my project, Online Learning Platform. The main purpose of this project is to provide an online learning environment where students can learn courses, instructors can share educational content, and administrators can manage the platform. For this project, I designed a MySQL database named online_learning with

15 tables. These tables are connected using primary keys and foreign keys. The database stores users, courses, lessons, enrollments, payments, quizzes, live classes, certificates, reviews and wishlists. I plan to develop the application using PHP and MySQL. Thank you.

For this project, I designed a MySQL database named online_learning with 15 tables. The users table stores admin, instructor and student information. The courses table stores course details. Course sections and lessons organize the learning materials. Enrollments and payments store student course and payment records. The quiz system contains quizzes, questions, options and quiz attempts. The database also includes live classes, certificates, course reviews and wishlists.

These tables are connected using primary keys and foreign keys. I plan to develop the application using PHP and MySQL. Thank you.

Database Reference Note

This document follows the structure of the SQL dump supplied for the project. The supplied database is named online_learning and contains 15 tables. Field names and table names in this document follow that SQL file. The SkillSet application is used as a functional reference; this document does not claim to reproduce its private original database.